with redundant subsystems, and operated by dedicated and experienced engineers, but they are not perfect. The amazing thing is not that we have done so poorly, but that we have done so well.

We don't know whether the *Mars Observer* failure was due to incompetence or just statistics. But we must expect a steady background of mission failures when we explore other worlds. No human lives are risked when a robot spacecraft is lost. Even if we were able to improve this success rate significantly, it would be far too costly. It is much better to take more risks and fly more spacecraft.

Knowing about irreducible risks, why do we these days fly only one spacecraft per mission? In 1962 *Mariner 1*, intended for Venus, fell into the Atlantic; the nearly identical *Mariner 2* became the human species' first successful planetary mission. *Mariner 3* failed, and its twin *Mariner 4* became, in 1964, the first spacecraft to take close-up pictures of Mars. Or consider the 1971 *Mariner 8/Mariner 9* dual launch mission to Mars. *Mariner 8* was to map the planet. *Mariner 9* was to study the enigmatic seasonal and secular changes of surface markings. The spacecraft were otherwise identical. *Mariner 8* fell into the ocean. *Mariner 9* flew on to Mars and became the first spacecraft in human history to orbit another planet. It discovered the volcanos, the laminated terrain in the polar caps, the ancient river valleys, and the aeolian nature of the surface changes. It disproved the "canals." It mapped the planet pole to pole and revealed all the major geological features of Mars known to us today. It provided the first close-up observations of members of a whole class of small worlds (by targeting the Martian moons, Phobos and Deimos). If we had launched only *Mariner 8*, the endeavor would have been an unmitigated failure. With a dual launch it became a brilliant and historic success.

There were also two *Vikings*, two *Voyagers*, two *Vegas*, many pairs of *Veneras*. Why was only one *Mars Observer* flown? The standard answer is cost. Part of the reason it was so costly, though, is that it was planned to be launched by shuttle, which is an almost absurdly expensive booster for planetary missions—in this case too expensive for two *M. O.* launches. After many shuttle-connected delays and